Inter - Conversion of Units

1 Å = 10^{-10} m

1 atm = 101.325 kPa = 1.01325 x 10^5 N m^{-2}

1 atm = 1.01325 x 10^5 J m^{-3} = 760 Torr (mmHg)

1 eV = 1.60219 x 10^{-19} J

1 N = 1 J m^{-1} = 10^5 dyne

1 J = 10^7 erg = 1 Kg m^2 s^{-2}

1 cal = 4.184 J

Critical constants of selected gases

Gas	Critical Pressure, P_c (atm)	Critical Volume, V_c (cm^3 mol^{-1})	Critical Temperature, T_c (K)
He	2.26	57.76	5.21
Ne	26.86	41.74	44.44
Ar	48.00	75.25	150.72
Kr	54.27	92.24	209.39
Xe	58.0	118.8	289.75
H_2	12.8	64.99	33.23
N_2	33.54	90.10	126.3
O_2	50.14	78.0	154.8
Cl_2	76.1	124	417.2
Br_2	102	135	584
HCl	81.5	81.0	324.7
CO_2	72.85	94.0	304.2
H_2O	218.3	55.3	647.4
NH_3	111.3	72.5	405.5
CH_4	45.6	98.7	190.6
C_2H_4	50.50	124	283.1
C_2H_6	48.20	148	305.4
C_6H_6	48.6	260	562.7

11th Std Chemistry 229-289.indd 231 06-01-2021 16:04:23